Form Controls Documentation

Integrated Customer Relationship Management and Production Assigning and Monitoring System (CRM-PAMS), CADDSolutions Incorporated, Cubao Branch

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1. Overview

This document describes the forms built for the Form Controls activity of the final project. The project is a Java SE 17 desktop application with a Java Swing interface and a MySQL 8.0 database, as stated in Chapters 1 and 3. Swing controls are used as the equivalents of HTML form controls, as the activity allows.

Form	Activity type	Module (Chapter 2 numbering)	Users
New Order	Major form	5.0 Customer Order Intake (IPO 5.1 to 5.4)	Operations Manager, Officer-in-Charge
Register New Employee	Major form	11.0 Registration (IPO 11.2)	Operations Manager
Search Orders	Supporting form (search)	12.0 Search (IPO 12.1 to 12.3)	Operations Manager, Officer-in-Charge
Production Monitoring	Supporting form (filter)	9.0 Production Status Monitoring (IPO 9.1 to 9.4)	Operations Manager, Officer-in-Charge

All four forms follow the same design:

- **Labels.** Every input has a visible label linked to it with `setLabelFor`, and most labels have an Alt key shortcut (the underlined letter).
- **Placeholder and helper text.** Text fields show a gray example while empty, and a short helper line under the input explains the expected format.
- **Validation in two layers.** The form checks its input before sending it. The service class checks it again, together with database checks such as unique usernames and duplicate orders. Both layers call the same rules in `validation/FieldRules.java`, so they cannot disagree.

- **Error messages.** On a failed submit, nothing is saved. Each invalid field gets a red border and an “Error:” line under it, a summary of all problems appears at the top, and the cursor moves to the first invalid field. Color is never the only signal.
- **Keyboard use.** Tab moves through the controls from top to bottom, then to the buttons, then to the results table. Enter runs the main button and Escape runs Cancel or Reset. Radio buttons change with the arrow keys, and Tab leaves text areas instead of typing a tab.
- **Screen readers.** Every input has an accessible name (“Customer name, required”) and an accessible description (the helper text, or the error when there is one).
- **Data handling.** Every form uses real data: two forms save to MySQL, and two run queries and show the results in a table.

2. Major Form 1: New Order

Purpose. Records a customer order once, in one place, instead of copying it from Facebook, Viber, Gmail, or phone notes into a Google Sheet and a logbook (Chapter 1, Specific Problem 1 and Objective 1). The order is saved with the status New and becomes available to the other modules.

Module. Customer Order Intake (5.0): Customer Lookup (5.1), New Customer Registration (5.2), Order Details Entry and Validation (5.3), and Order Record Creation (5.4).

2.1 Controls (18)

#	Label	Control (Swing)	HTML equivalent	Required	Purpose
1	Customer name	`JTextField` with placeholder	Textbox	Yes	Person or company name
2	Contact number	`JTextField` that accepts digits and + only	Tel / textbox	Yes	Used with the name to find an existing customer
3	Email	`JTextField`	Email	No	Saved with a new customer record
4	Look Up Customer	`JButton`	Button		Finds an existing customer (IPO 5.1)
5	Project type	`JComboBox` loaded from `project_types`	Select	Yes	Kind of drafting or printing work
6	Project description	`JTextArea` with live character count	Textarea	Yes	What the customer needs
7	Required output	`JTextField`	Textbox	No	Number of sets, size, paper
8	Deadline	`JSpinner` with a date model (YYYY-MM-DD)	Date	Yes	Primary key of the production queue
9	Urgency level	Three `JRadioButtons` loaded from `urgency_levels`	Radio buttons	Yes	Secondary key of the production queue
10	Special instructions	`JTextArea`	Textarea	No	Paper type, sheet size, other notes

#	Label	Control (Swing)	HTML equivalent	Required	Purpose
11	Reference file	Read-only `JTextField`	Textbox (read-only)	No	Shows the chosen file path
12	Browse Reference File	`JButton` opening a `JFileChooser`	File upload	No	Picks a design or reference file
13	Remove File	`JButton`	Button		Removes the chosen file
14	Customer ID	Hidden value set by Look Up Customer	Hidden field		Links the order to the found customer
15	Encoded by	Hidden value from the logged-in user	Hidden field		Records who encoded the order
16	Save Order	`JButton` (Enter)	Submit button		Validates and saves
17	Clear Form	`JButton`	Reset button		Empties every field
18	Cancel	`JButton` (Escape)	Button		Closes the form, asking first if data was typed

The date received is not an input. The system sets it when the order is saved, so it cannot be typed wrongly.

2.2 Validation rules

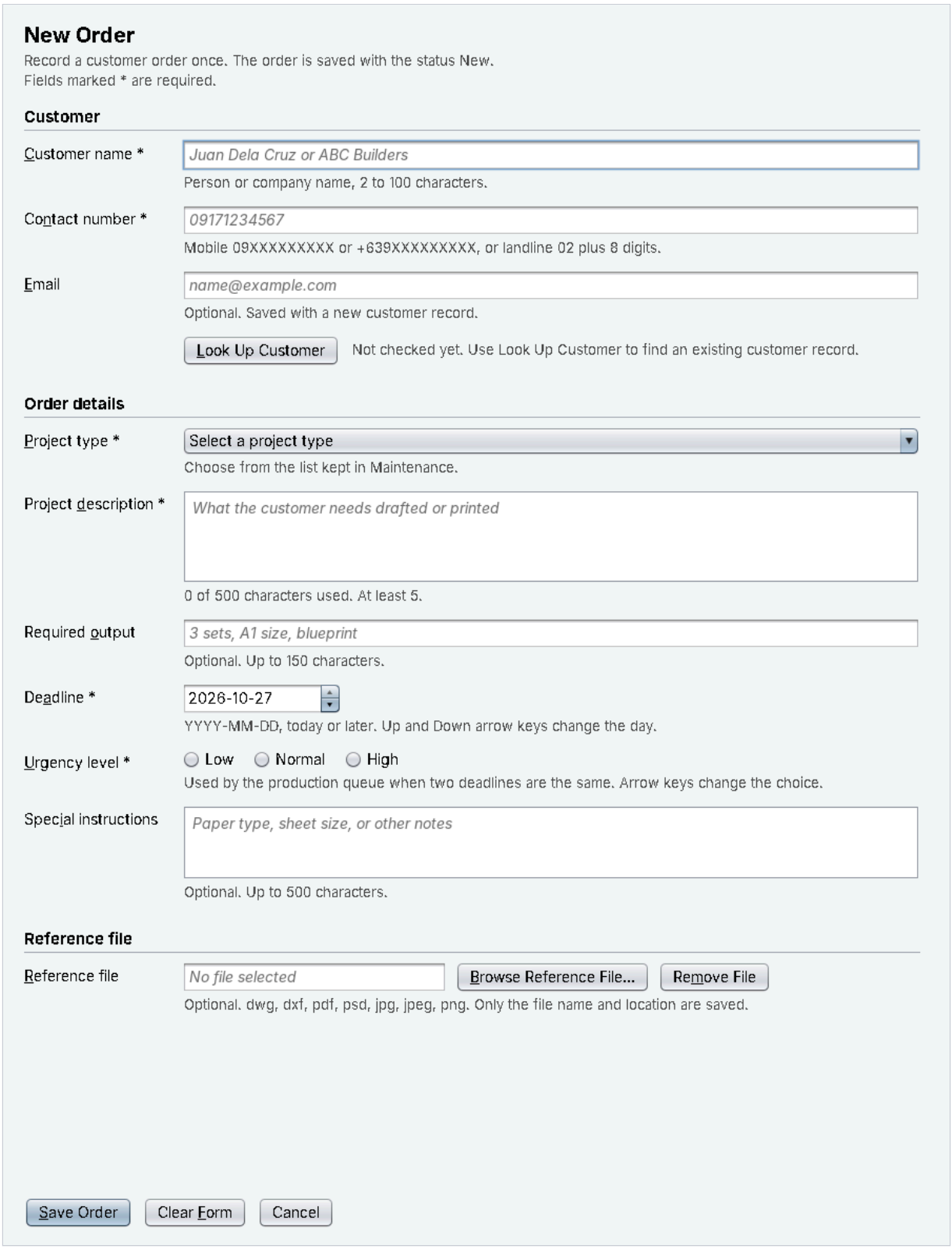
Field	Rule	Checked in	Message shown
Customer name	Required; 2 to 100 characters; letters, numbers, spaces, and . , ’ & ( ) - only; at least one letter	Form and service	“Customer name must be 2 to 100 characters long.”
Contact number	Required; 09 plus 9 digits, +639 plus 9 digits, or 02 plus 8 digits; stored as 09XXXXXXXXX	Typing filter, form, and service	“Enter a mobile number like 09171234567 or +639171234567, or a landline like 0281234567.”
Email	Optional; name@domain.tld pattern; up to 100 characters	Form and service	“Enter an email address like name@example.com.”
Project type	Required; must still be active in Maintenance	Form; service checks the database	“Choose a project type.”

Field	Rule	Checked in	Message shown
Project description	Required; 5 to 500 characters; typing stops at 500	Typing filter, form, and service	“Project description must be at least 5 characters long.”
Required output	Optional; up to 150 characters	Typing filter, form, and service	“Required output must be 150 characters or fewer.”
Deadline	Required; a real date in YYYY-MM-DD form; today or later	Form and service	“The deadline cannot be earlier than today.” or “Enter the deadline as YYYY-MM-DD.”
Urgency level	Required; must still be active	Form; service checks the database	“Choose an urgency level.”
Special instructions	Optional; up to 500 characters	Typing filter, form, and service	“Special instructions must be 500 characters or fewer.”
Reference file	Optional; dwg, dxf, pdf, psd, jpg, jpeg, or png; the file must exist; path up to 500 characters	Form and service	“The file was not found. Check that the shared folder is reachable and choose the file again.”
Customer record	A customer with the same name and contact number is reused, never created twice	Service	Shown on screen as the Look Up Customer result
Duplicate order	An open order of the same customer with the same description triggers a Yes or No question before saving	Service	“Order 1001 of this customer has the same project description and is still open. Save this as a separate new order anyway?”
Role	Only the Operations Manager and Officer-in-Charge may save orders	Service	“Your role (Production Staff) is not allowed to use this function.”

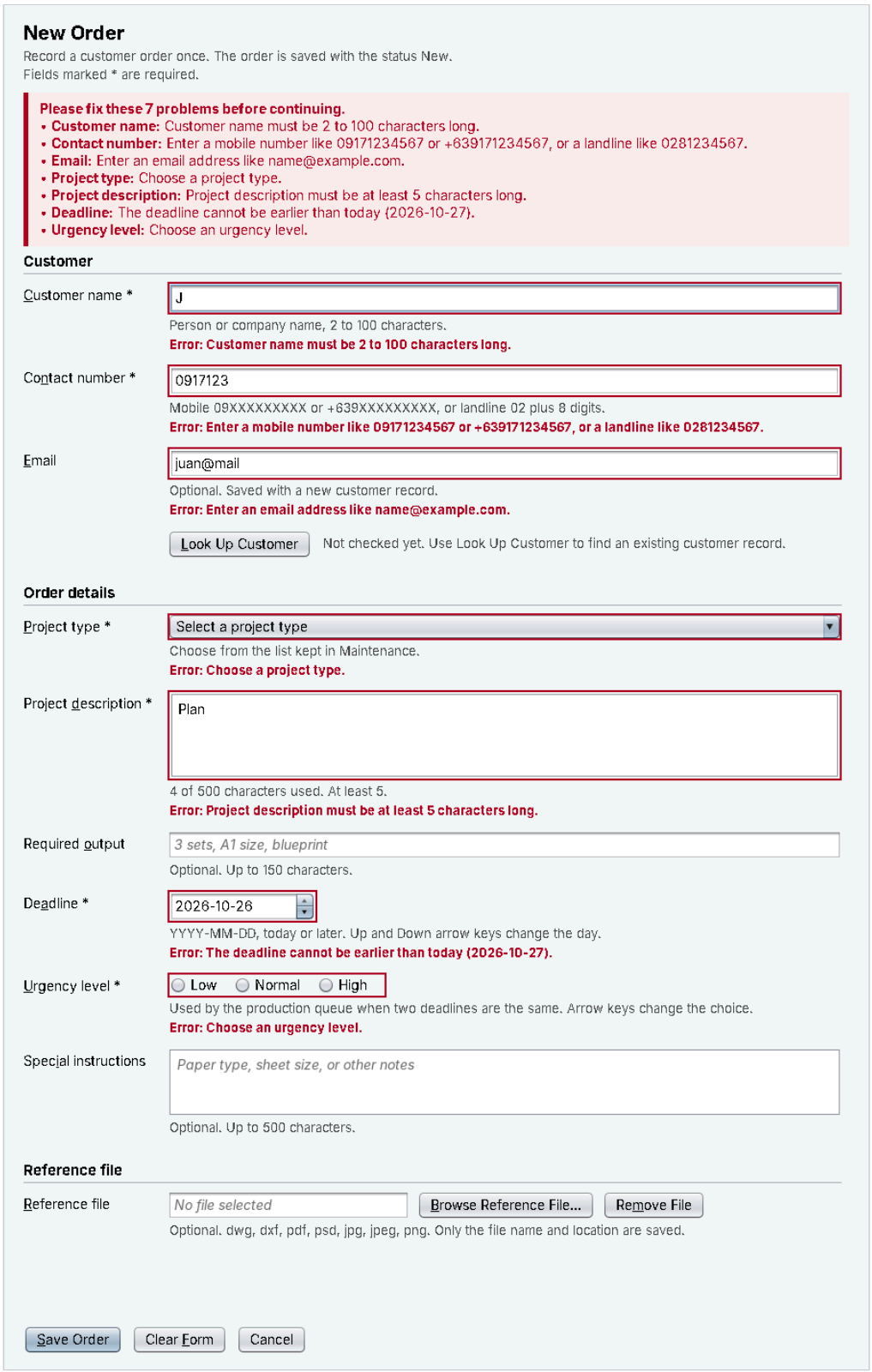
2.3 Data handling

`OrderIntakeService.save` runs one database transaction that inserts the customer (when new) into `customers`, the order into `orders` with the stage New, the reference file name and location into `reference_files`, a row in `status_history`, and rows in `activity_log`. If any step fails, the whole transaction is rolled back. All statements are JDBC prepared statements.

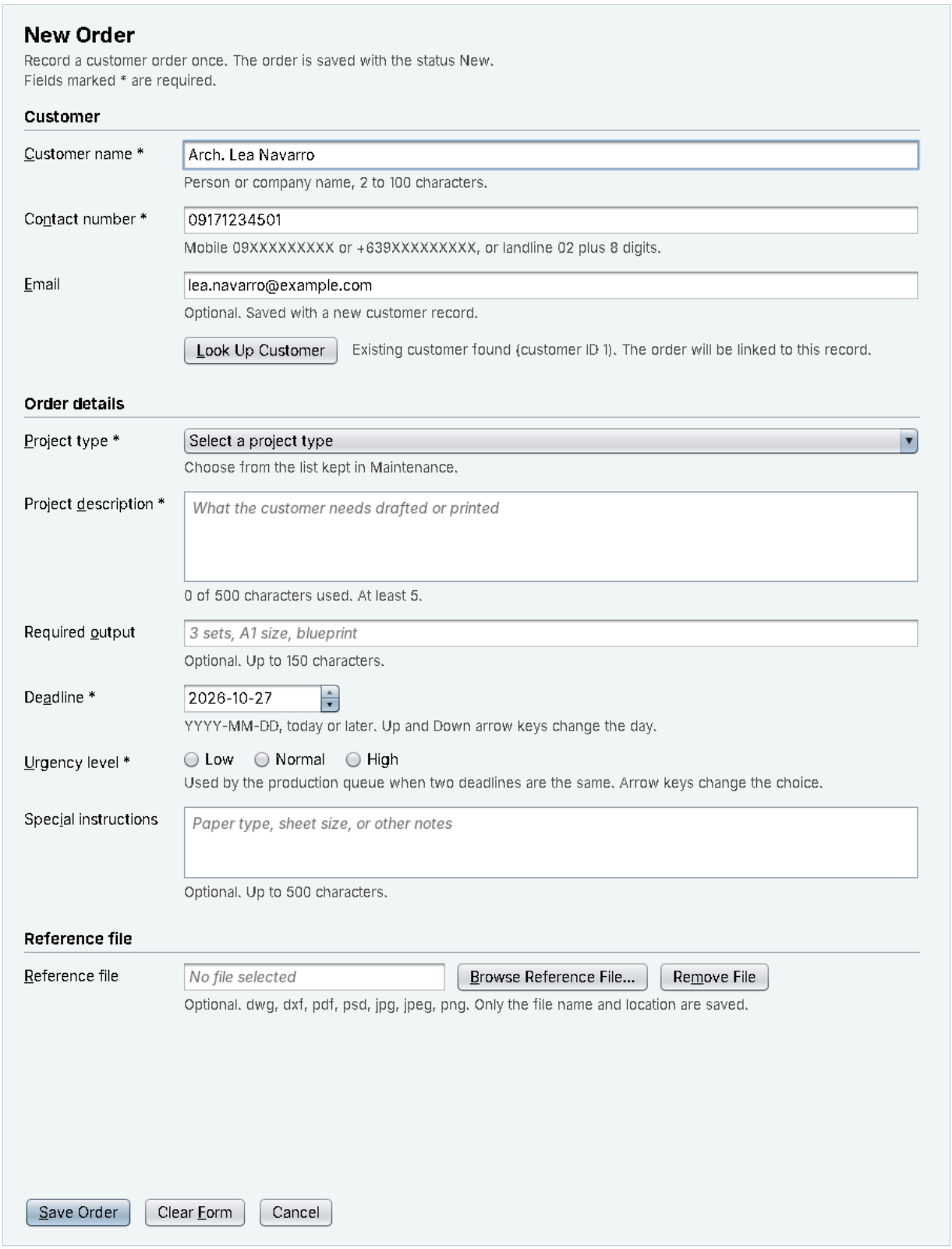
2.4 Screenshots



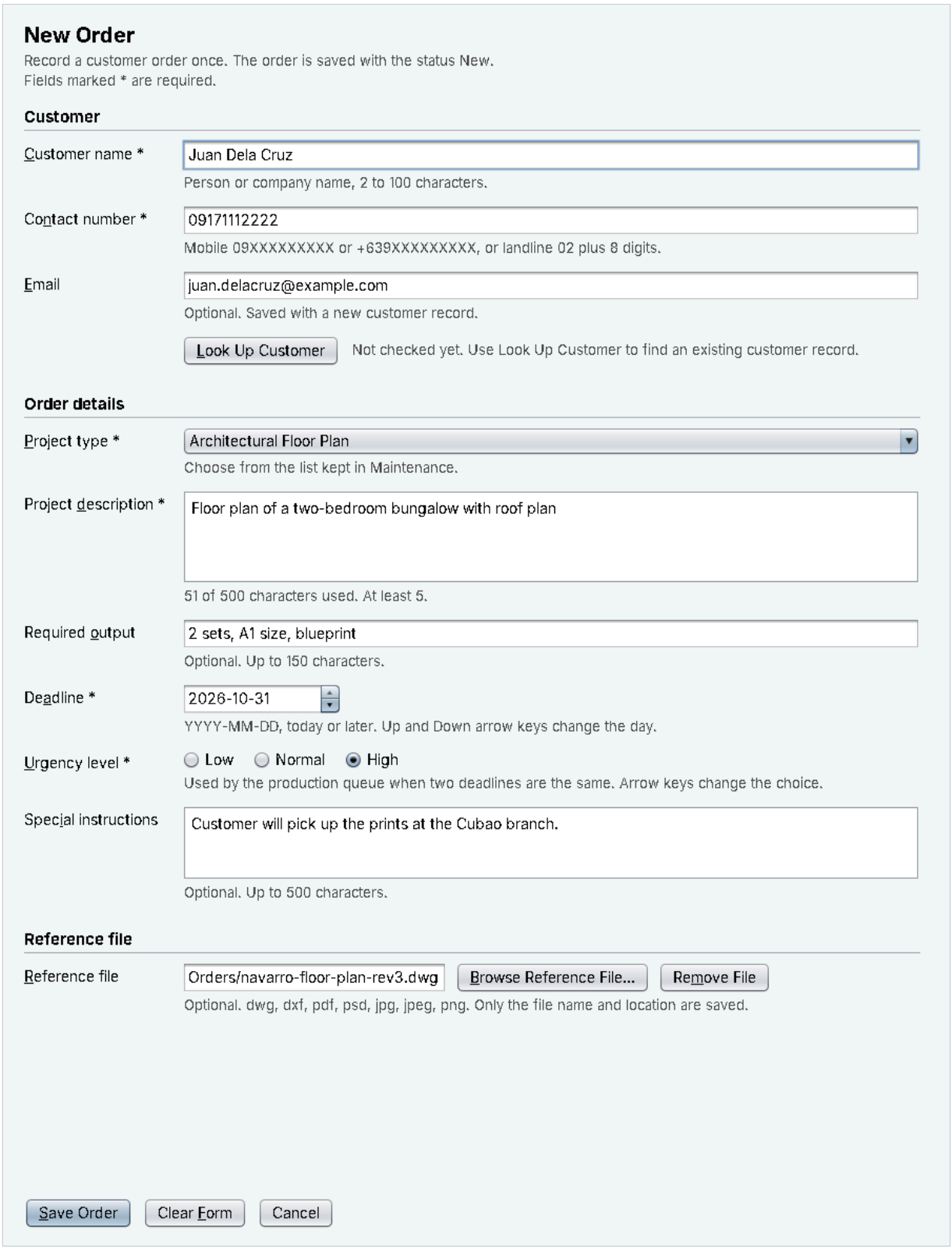
New Order form, normal state



New Order form, error state



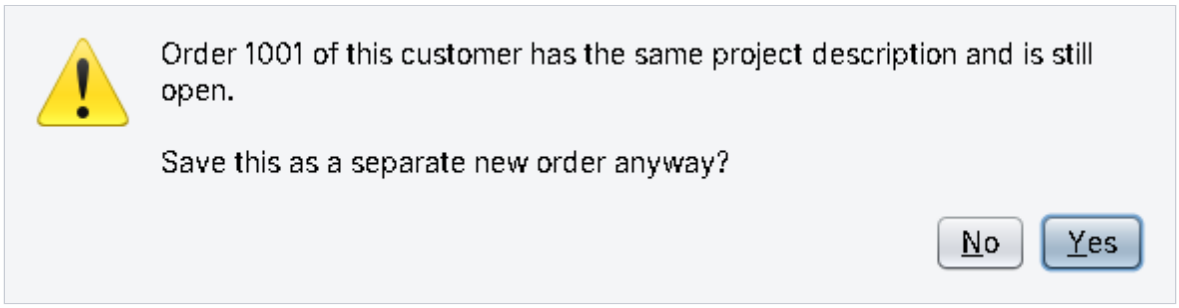
Look Up Customer found an existing customer



New Order form filled with valid data



Message after saving



Possible duplicate order warning

3. Major Form 2: Register New Employee

Purpose. Lets the Operations Manager create the login account of an employee, assign the role that decides which menu the employee sees, and set an initial password (Chapter 1, Registration module; Chapter 2, Figure 20).

Module. Registration (11.0), Register New Employee (IPO 11.2).

3.1 Controls (15)

#	Label	Control (Swing)	HTML equivalent	Required	Purpose
1	First name	`JTextField` with placeholder	Textbox	Yes	Employee's first name
2	Last name	`JTextField` with placeholder	Textbox	Yes	Employee's last name
3	Role	Three `JRadioButtons`	Radio buttons	Yes	Operations Manager, Officer-in-Charge, or Production Staff
4	Position	`JComboBox`, enabled only for Production Staff	Select	For Production Staff	Designer, Drafter, or Modeler
5	Contact number	`JTextField` that accepts digits and + only	Tel / textbox	No	Employee contact
6	Email	`JTextField`	Email	No	Employee email
7	Username	`JTextField` that accepts letters, digits, . and _	Textbox	Yes	Login name
8	Password	`JPasswordField`	Password	Yes	Initial password
9	Confirm password	`JPasswordField`	Password	Yes	Catches typing mistakes
10	Show passwords	`JCheckBox`	Checkbox		Shows the typed passwords
11	Account status	`ToggleSwitch` (a styled `JToggleButton`)	Switch / toggle		Active or Inactive; inactive accounts cannot log in

#	Label	Control (Swing)	HTML equivalent	Required	Purpose
12	Created by	Hidden value from the logged-in user	Hidden field		Records who created the account
13	Register Employee	`JButton` (Enter)	Submit button		Validates and saves
14	Clear Form	`JButton`	Reset button		Empties every field
15	Cancel	`JButton` (Escape)	Button		Closes the form, asking first if data was typed

3.2 Validation rules

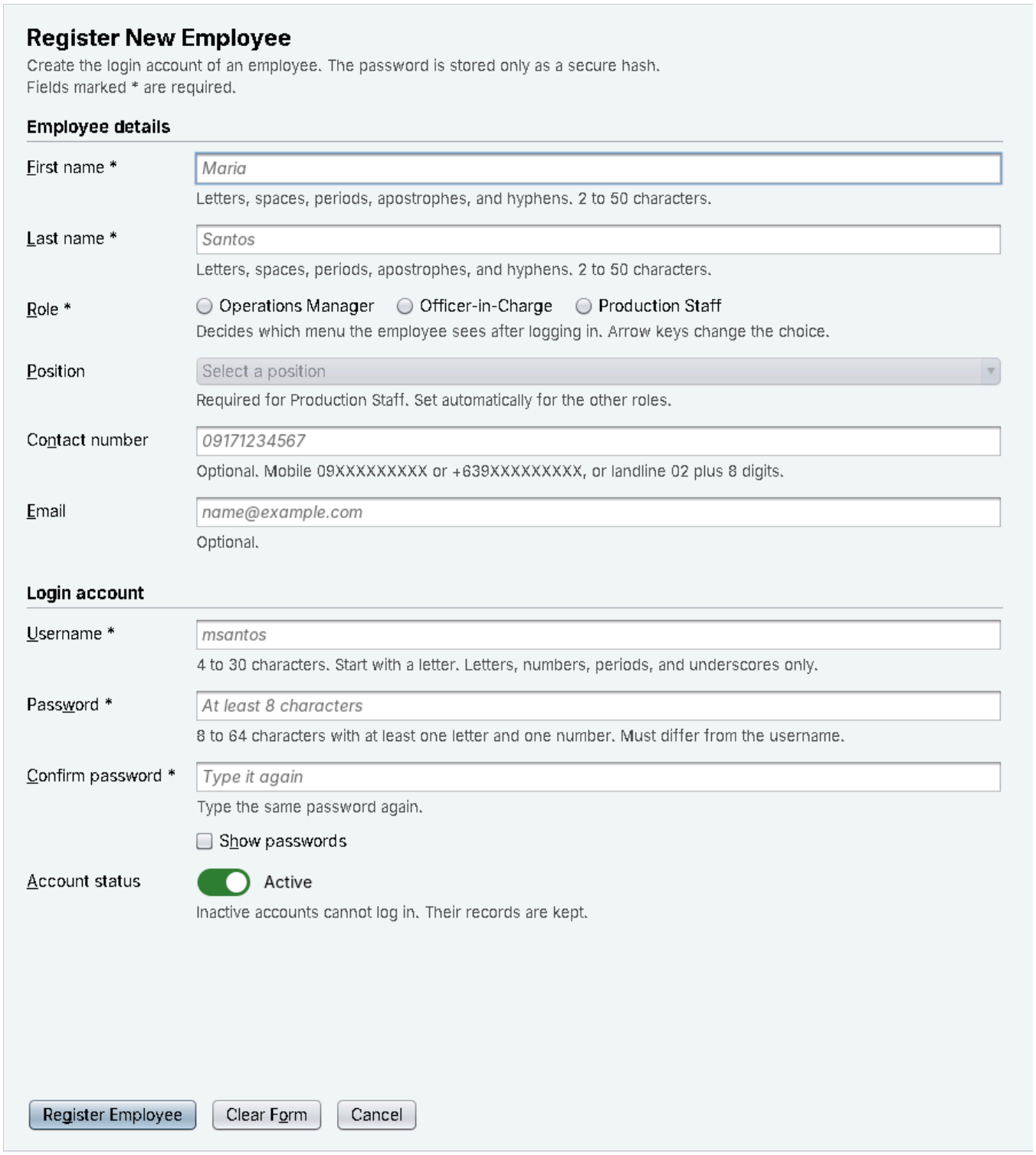
Field	Rule	Checked in	Message shown
First name, Last name	Required; 2 to 50 characters; letters, spaces, periods, apostrophes, and hyphens only	Typing filter, form, and service	“First name may use letters, spaces, periods, apostrophes, and hyphens only.”
Role	Required	Form and service	“Choose the role of this employee.”
Position	Required when the role is Production Staff; must be Designer, Drafter, or Modeler	Form and service	“Choose Designer, Drafter, or Modeler.”
Contact number	Optional; same formats as the New Order form	Typing filter, form, and service	“Enter a mobile number like 09171234567 ...”
Email	Optional; name@domain.tld pattern	Form and service	“Enter an email address like name@example.com.”
Username	Required; 4 to 30 characters; starts with a letter; letters, numbers, periods, and underscores	Typing filter, form, and service	“Username must be 4 to 30 characters long.”
Username	Must not already exist (letter case ignored); checked again inside the save transaction	Service (database)	“This username is already taken. Choose another one.”
Password	Required; 8 to 64 characters; at least one letter and one number; not the same as the username	Form and service	“Password must contain at least one letter and one number.”

Field	Rule	Checked in	Message shown
Confirm password	Must match the password	Form and service	“The two passwords do not match.”
Role of the user	Only the Operations Manager may register accounts	Service	“Your role (Officer-in-Charge) is not allowed to use this function.”

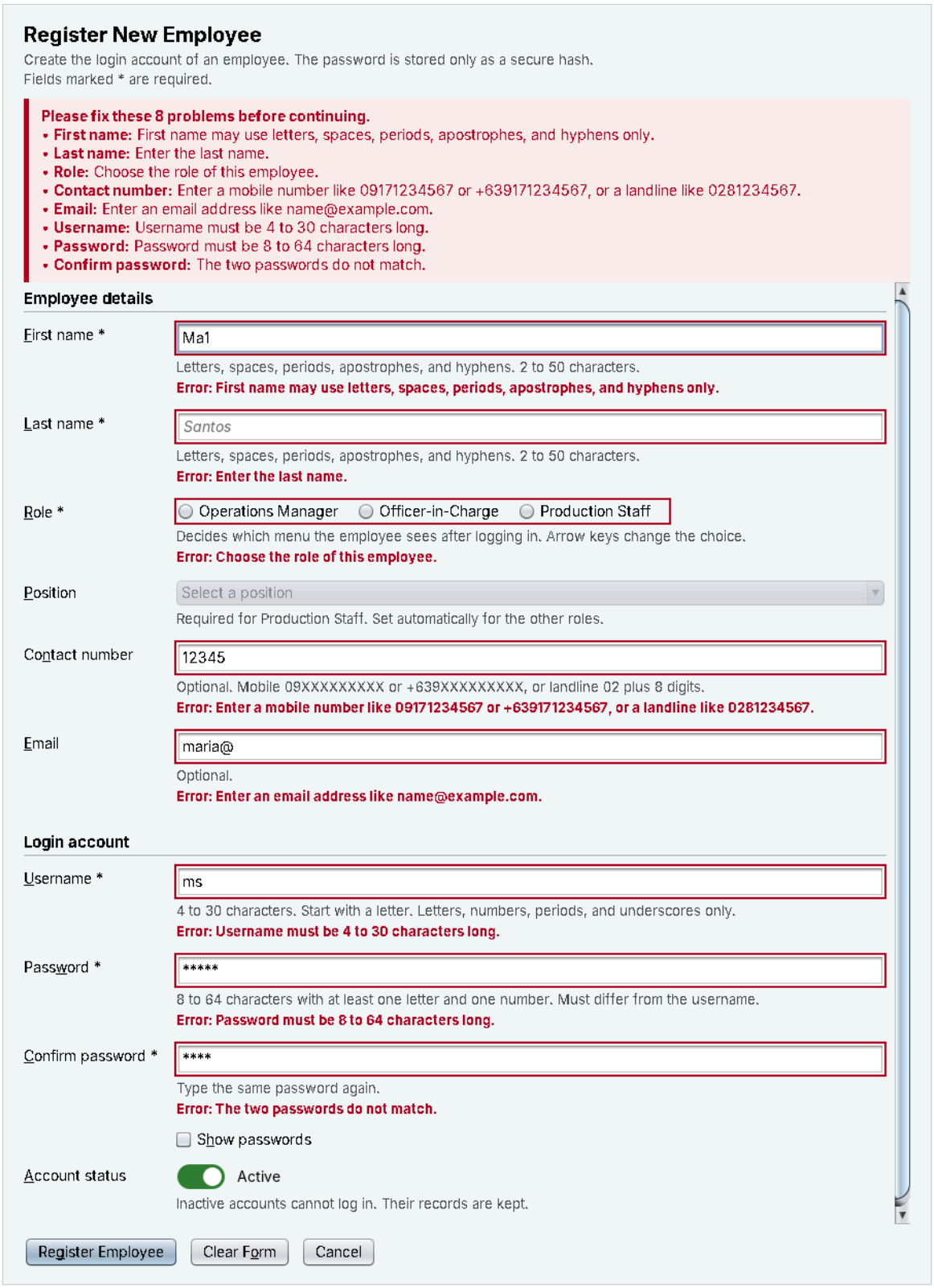
3.3 Data handling

The password is hashed with PBKDF2-HMAC-SHA256 at 600,000 iterations with a random salt for each user (Chapter 3, System Architecture) before it reaches the database. The plain password is never stored or logged, and the form clears the password arrays after use. `RegistrationService` inserts the employee into `employees` and the account into `users` in one transaction and adds a row to `activity_log`.

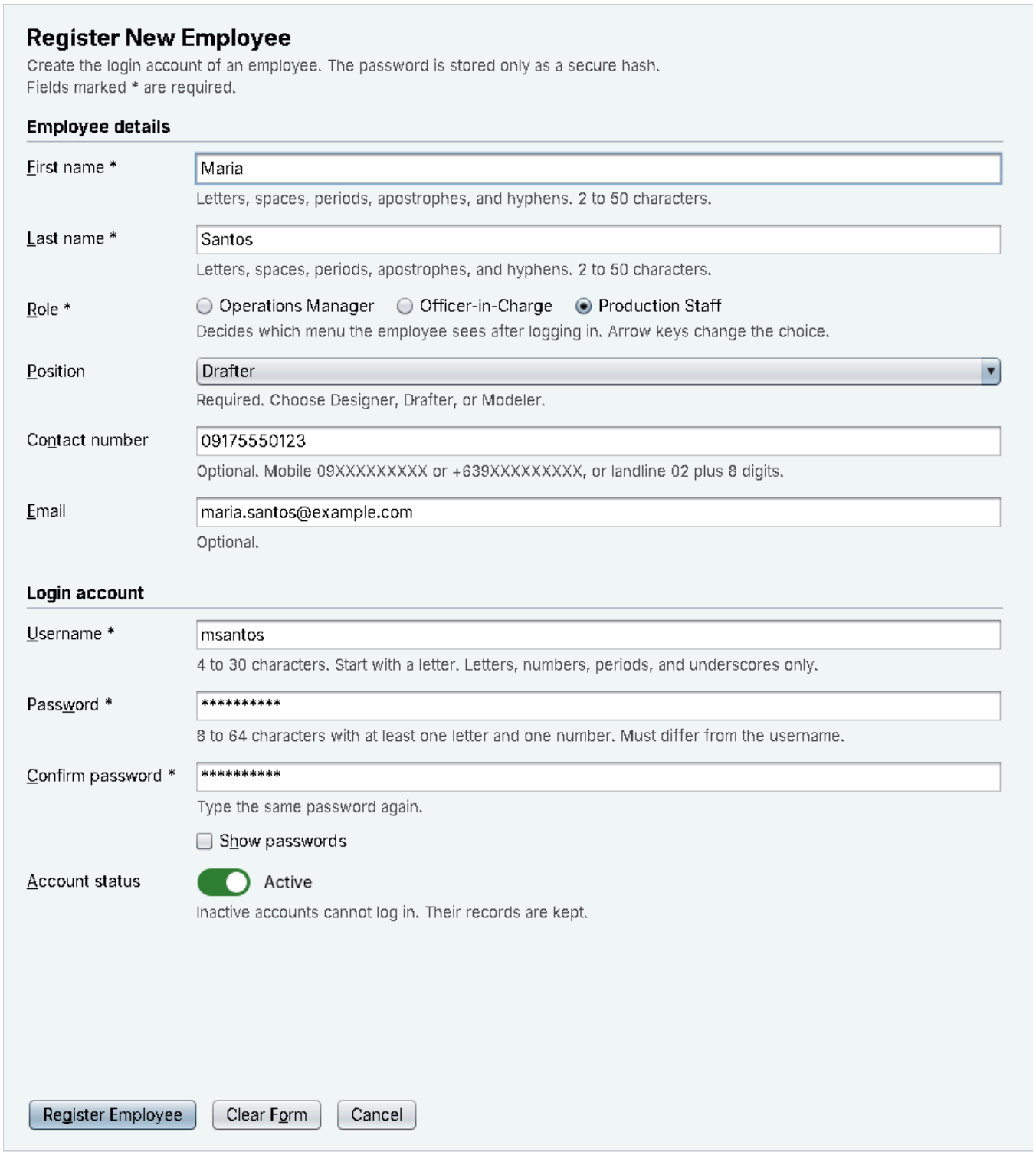
3.4 Screenshots



Register New Employee, normal state



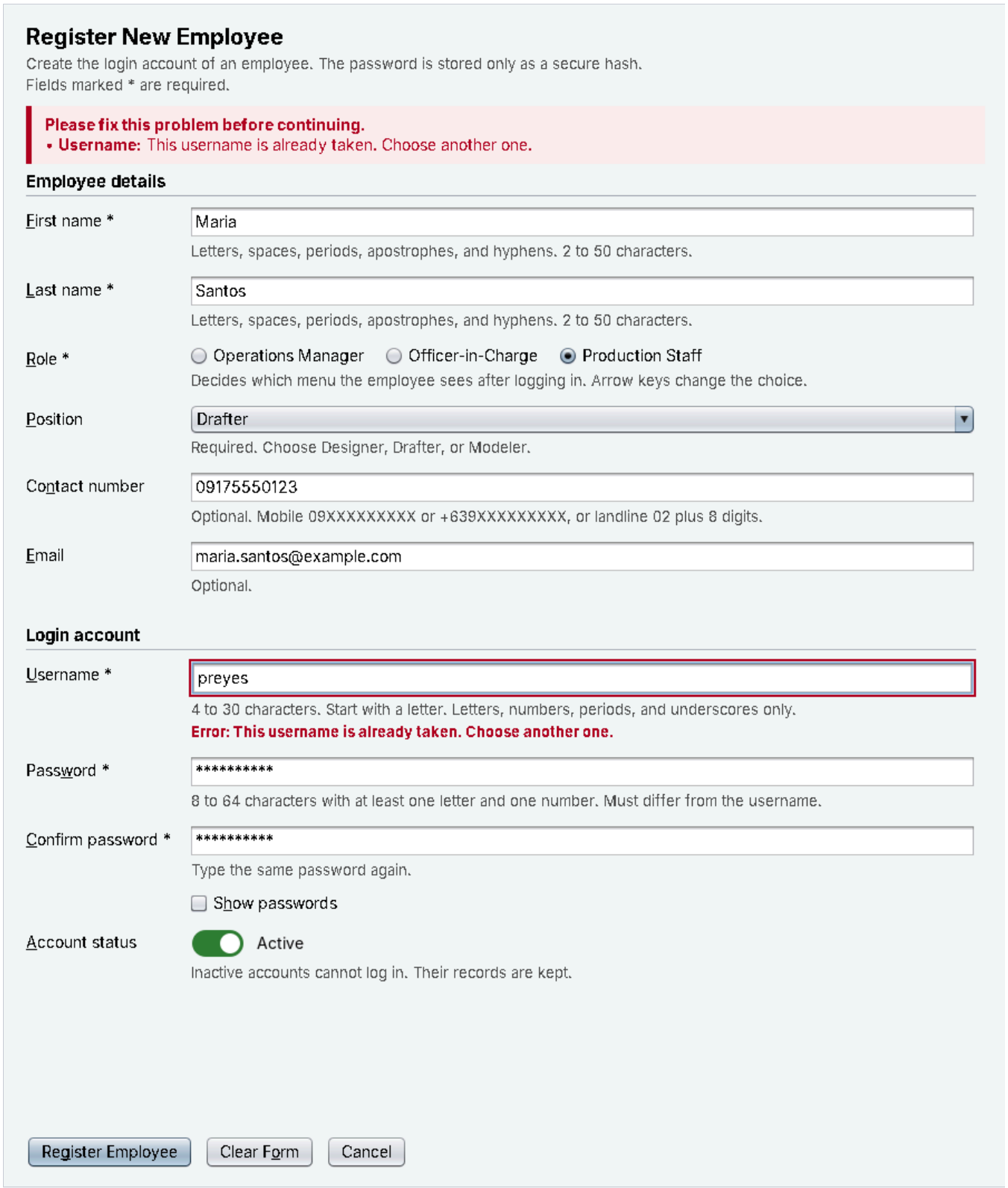
Register New Employee, error state



Register New Employee filled with valid data



Message after registering



Username already taken (server-side check)

4. Supporting Form 1: Search Orders

Purpose. Finds past and current orders without scrolling through a spreadsheet or turning the pages of a logbook (Chapter 1, Search module).

Module. Search (12.0): Enter Search Criteria (12.1), Run Search Query (12.2), and View Selected Order Record (12.3).

4.1 Controls (13)

#	Label	Control (Swing)	HTML equivalent	Purpose
1	Search text	`JTextField` with placeholder	Search box	Partial match on order ID, customer name, or description
2	Project type	`JComboBox` with “All project types”	Select	Filter
3	Status	`JComboBox` with “All statuses”	Select	Filter by workflow stage
4	Assigned employee	`JComboBox` with “All employees”	Select	Filter by current assignee
5	Urgency level	`JComboBox` with “All urgency levels”	Select	Filter (the “priority level” filter of Chapter 1)
6	Include released and completed orders	`JCheckBox`	Checkbox	Closed orders are hidden unless ticked
7	Filter by date received	`JCheckBox`	Checkbox	Turns the date range on or off
8	Received from	`JSpinner` date	Date	Start of the range
9	Received to	`JSpinner` date	Date	End of the range
10	Search Orders	`JButton` (Enter)	Submit button	Runs the search
11	Reset Filters	`JButton` (Escape)	Reset button	Clears every filter
12	Results	`JTable`, sortable by column	Table	Shows the matching orders
13	View Selected Order	`JButton` (also Enter or double-click on a row)	Button	Opens the full order record (IPO 12.3)

All filters are optional. Leaving everything blank lists the open orders.

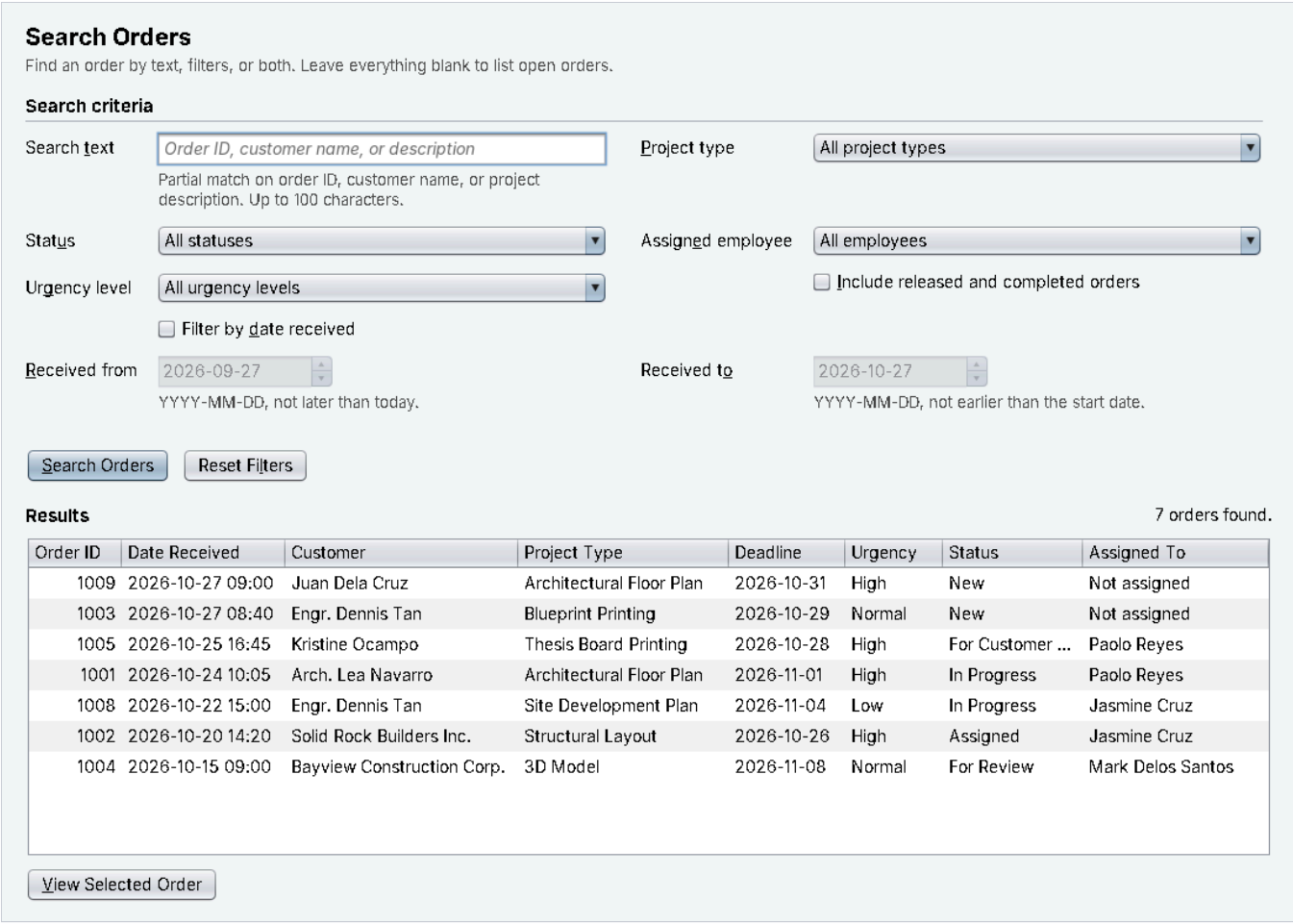
4.2 Validation rules

Field	Rule	Checked in	Message shown
Search text	Up to 100 characters	Typing filter, form, and service	“Search text must be 100 characters or fewer.”
Received from	When the date filter is on: a real date, not later than today	Form and service	“The start date cannot be later than today.”
Received to	When the date filter is on: a real date, not earlier than the start date	Form and service	“The end date cannot be earlier than the start date.”
Status	Choosing Released or Completed requires the Include released and completed orders box	Form and service	“To search Completed orders, also tick Include released and completed orders.”
Role	Operations Manager and Officer-in-Charge only	Service	“Your role (Production Staff) is not allowed to use this function.”

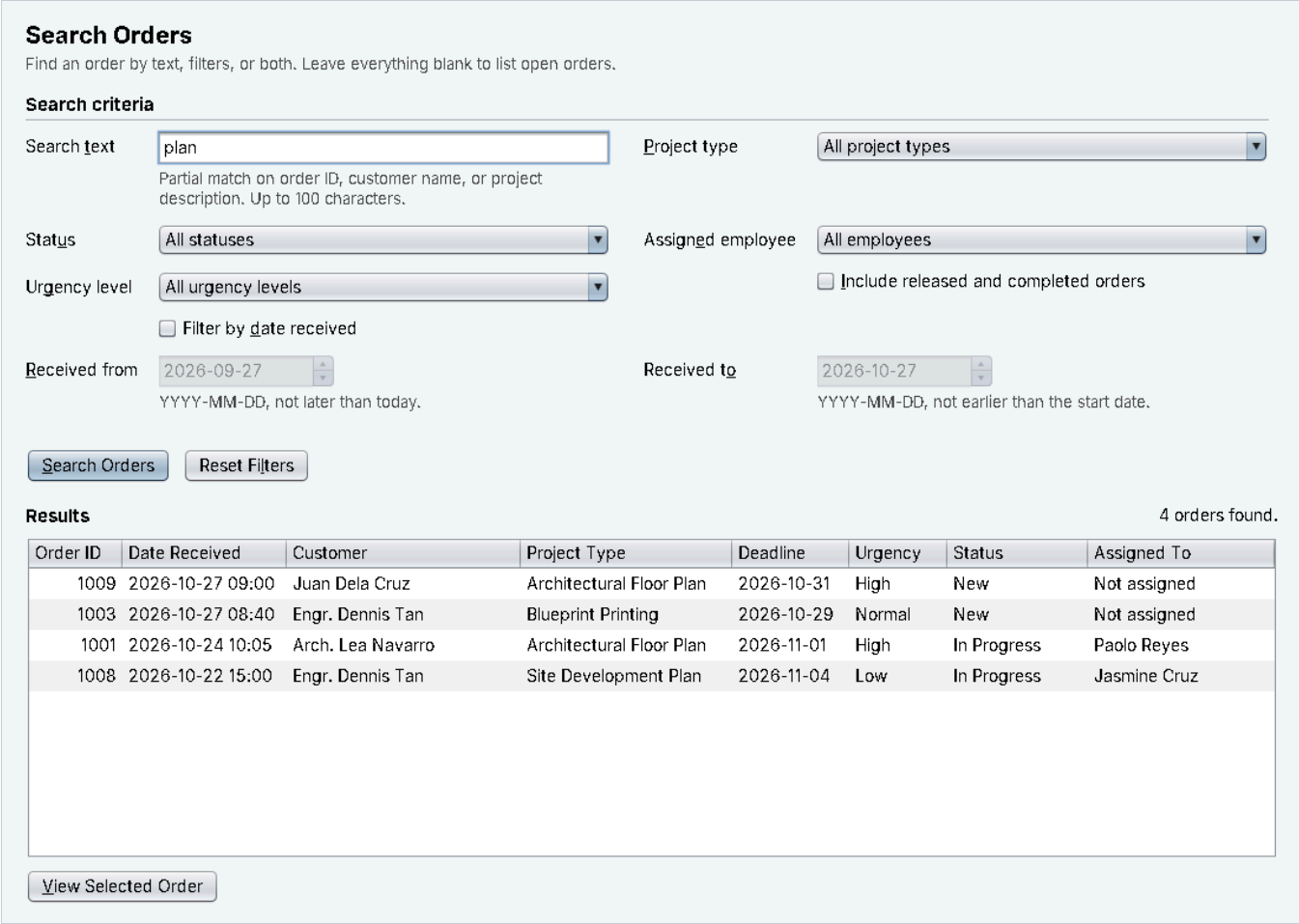
4.3 Data handling

`SearchService` turns the criteria into one parameterized SQL query (Chapter 3, Algorithm Implementation). Percent signs and underscores typed by the user are escaped, so they are matched as ordinary characters. The date range covers whole days from the start date to the end date. At most 500 rows are returned, newest first, and the count is shown above the table.

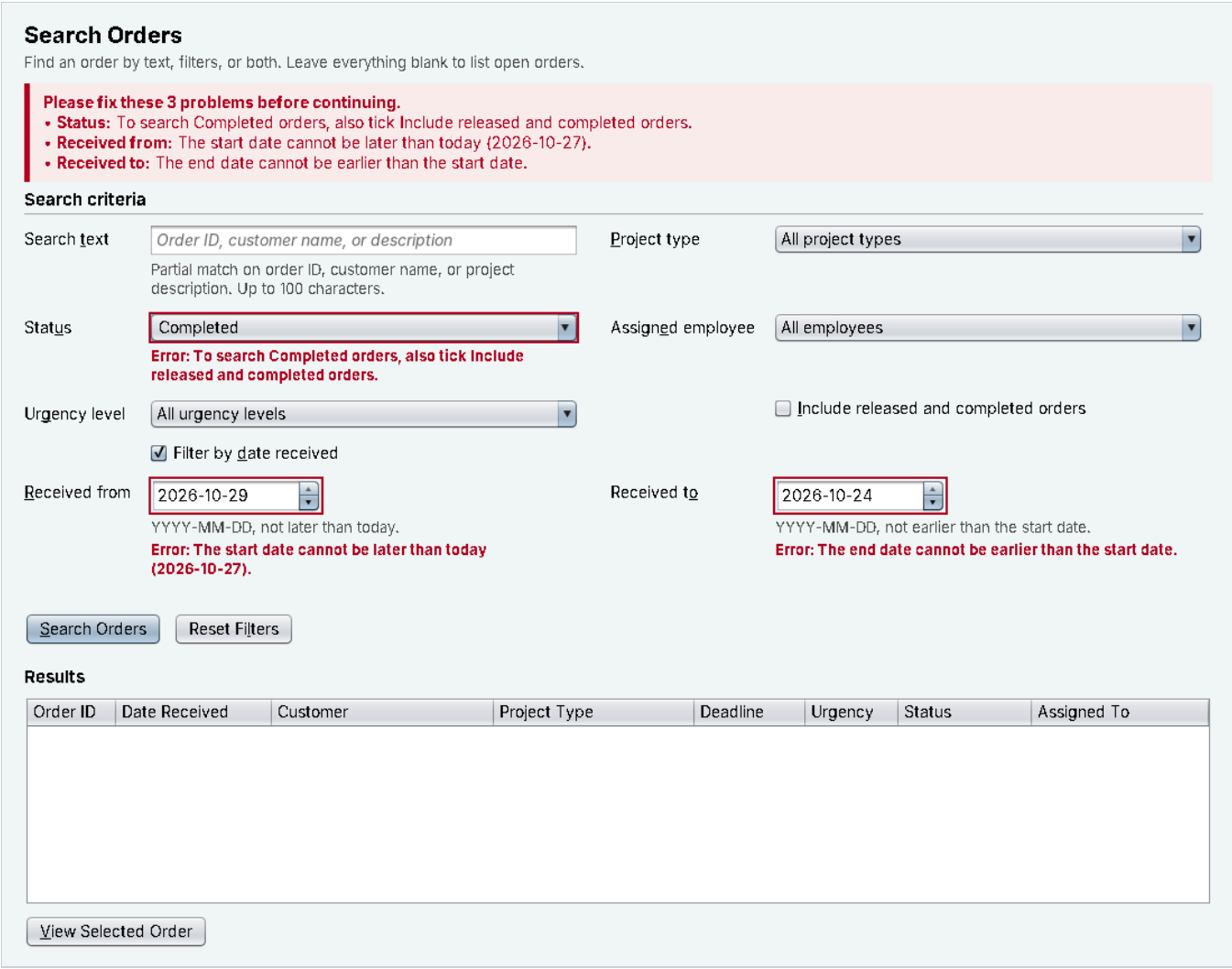
4.4 Screenshots



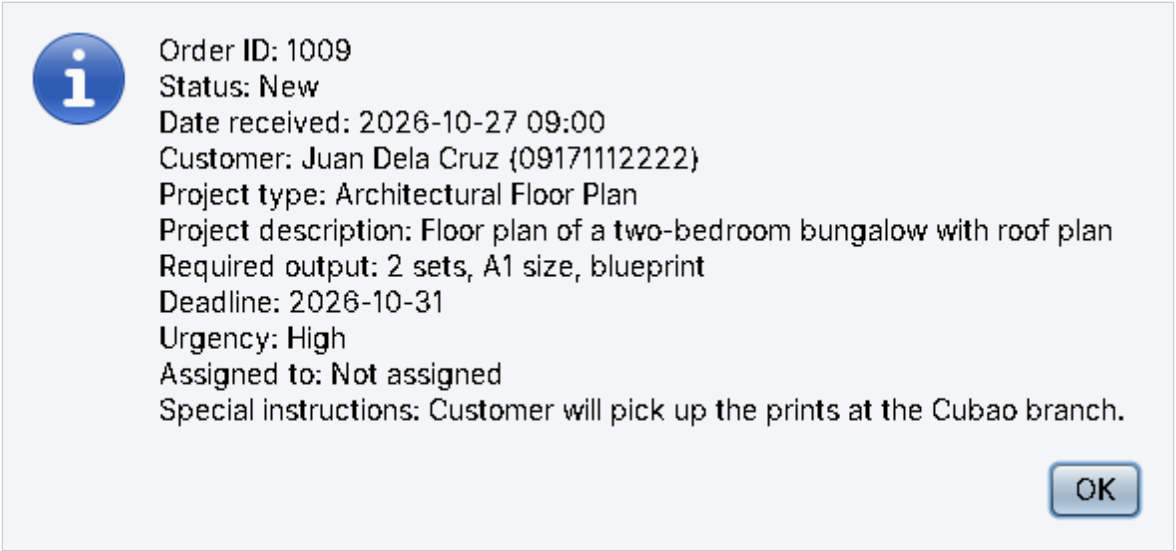
Search Orders listing open orders



Search for the text “plan”



Search Orders, error state



Viewing a selected order

5. Supporting Form 2: Production Monitoring (Monitoring Filter)

Purpose. Shows the stage, assignee, and deadline indicator of every active order so the Operations Manager and Officer-in-Charge no longer need to ask the assigned employee (Chapter 1, Specific Problem 4 and Objective 4).

Module. Production Status Monitoring (9.0): Management Monitoring View (9.1), Active Orders by Stage (9.2), Status Dashboard (9.3), and Deadline Indicators (9.4).

5.1 Controls (15)

#	Label	Control (Swing)	HTML equivalent	Purpose
1	Order ID or employee	`JTextField` with placeholder	Search box	Exact order ID, or part of an employee name (IPO 9.3)
2	Assigned employee	`JComboBox` with “All employees”	Select	Filter
3 to 7	Workflow stages	Five `JCheckBoxes`: New, Assigned, In Progress, For Review, For Customer Confirmation	Checkboxes	Stages to show (IPO 9.2)
8 to 11	Deadline status	Four `JRadioButtons`: All, Passed Deadline, Approaching Deadline, On Schedule	Radio buttons	Filter by indicator (IPO 9.4)
12	Warning period	`JSlider` from 1 to 14 days with the value shown beside it	Range slider	The warning period entered by the user (IPO 9.4)
13	Apply Filters	`JButton` (Enter)	Submit button	Refreshes the table
14	Reset Filters	`JButton` (Escape)	Reset button	Restores the default filters
15	Active orders	`JTable`, sortable, with the deadline status as text on a tinted cell	Table	Results, with a count per stage above it

5.2 Validation rules

Field	Rule	Checked in	Message shown
Order ID or employee	Up to 60 characters	Typing filter, form, and service	“Order ID or employee name must be 60 characters or fewer.”
Workflow stages	At least one stage ticked	Form and service	“Tick at least one workflow stage.”
Workflow stages	Released and Completed are ignored even if sent, because they are not active	Service	Not shown; closed orders never appear
Warning period	1 to 14 days (the slider cannot go outside this range, and the service checks again)	Slider limits and service	“Warning period must be 1 to 14 days.”
Role	Operations Manager and Officer-in-Charge only	Service	“Your role (Production Staff) is not allowed to use this function.”

5.3 Data handling

`MonitoringService` reads the active orders with their current assignee and last status update. For each order it computes the days remaining as the deadline minus the current date and applies the rule in Chapter 3: below zero is Passed Deadline, within the warning period is Approaching Deadline, and anything else is On Schedule. The form then shows the count of orders in each stage.

5.4 Screenshots

Production Monitoring

See the stage and deadline of every active order without asking the assigned employee. Click Apply Filters to refresh.
Fields marked * are required.

Filters

Order ID or employee: *Order ID or employee name*
Type an order ID for an exact match or part of an employee name. Leave blank for all.

Assigned employee: All employees

Workflow stages *:
- [x] New
- [x] Assigned
- [x] In Progress
- [x] For Review
- [x] For Customer Confirmation

Tick at least one. Released and Completed orders are not active.

Deadline status:
- (•) All
- ( ) Passed Deadline
- ( ) Approaching Deadline
- ( ) On Schedule

Arrow keys change the choice.

Warning period *: 3 days (1 … 7 … 14)
Orders due within this many days are marked Approaching Deadline. Arrow keys change it.

[Apply Filters] [Reset Filters]

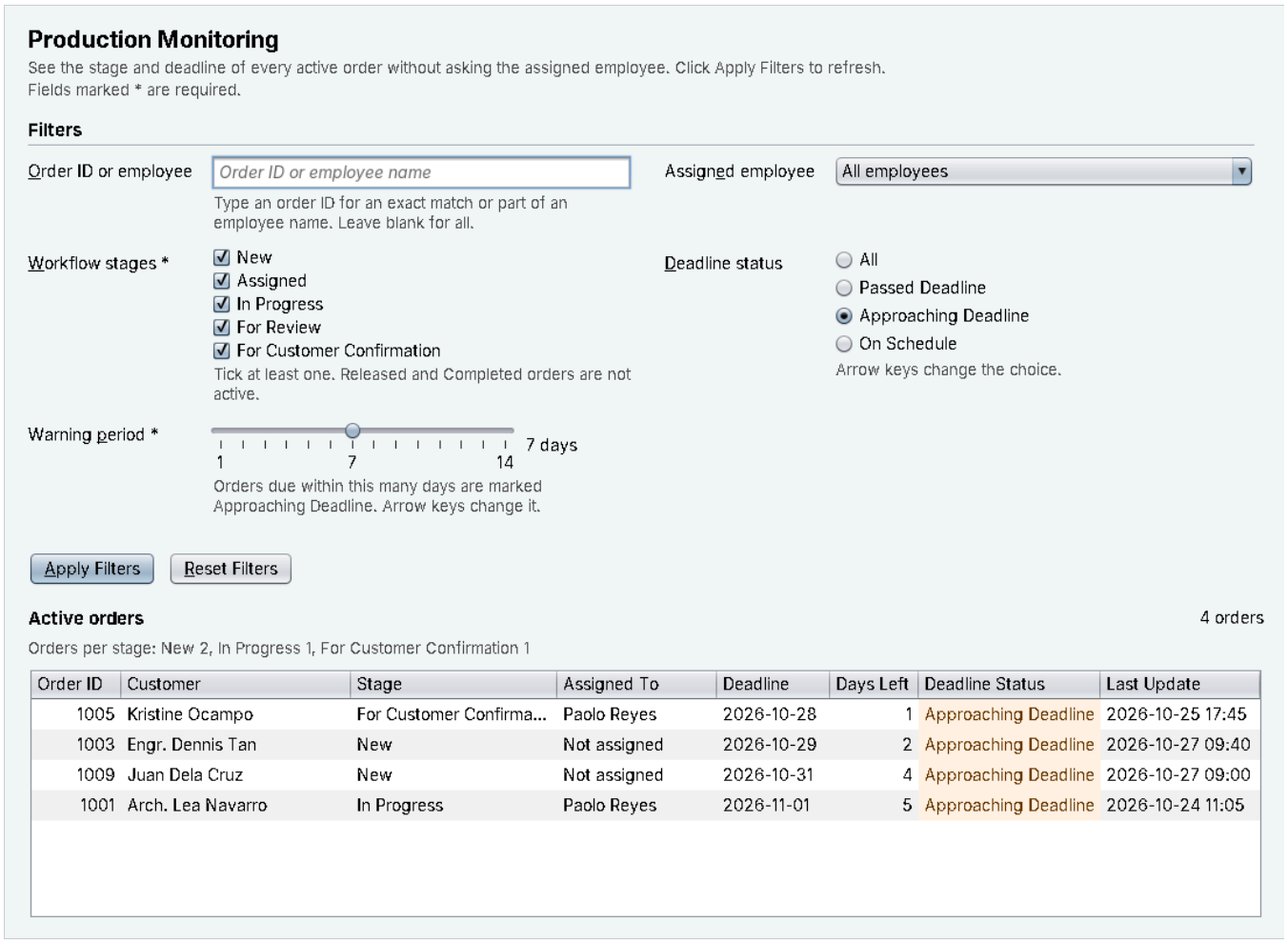
Active orders

7 orders

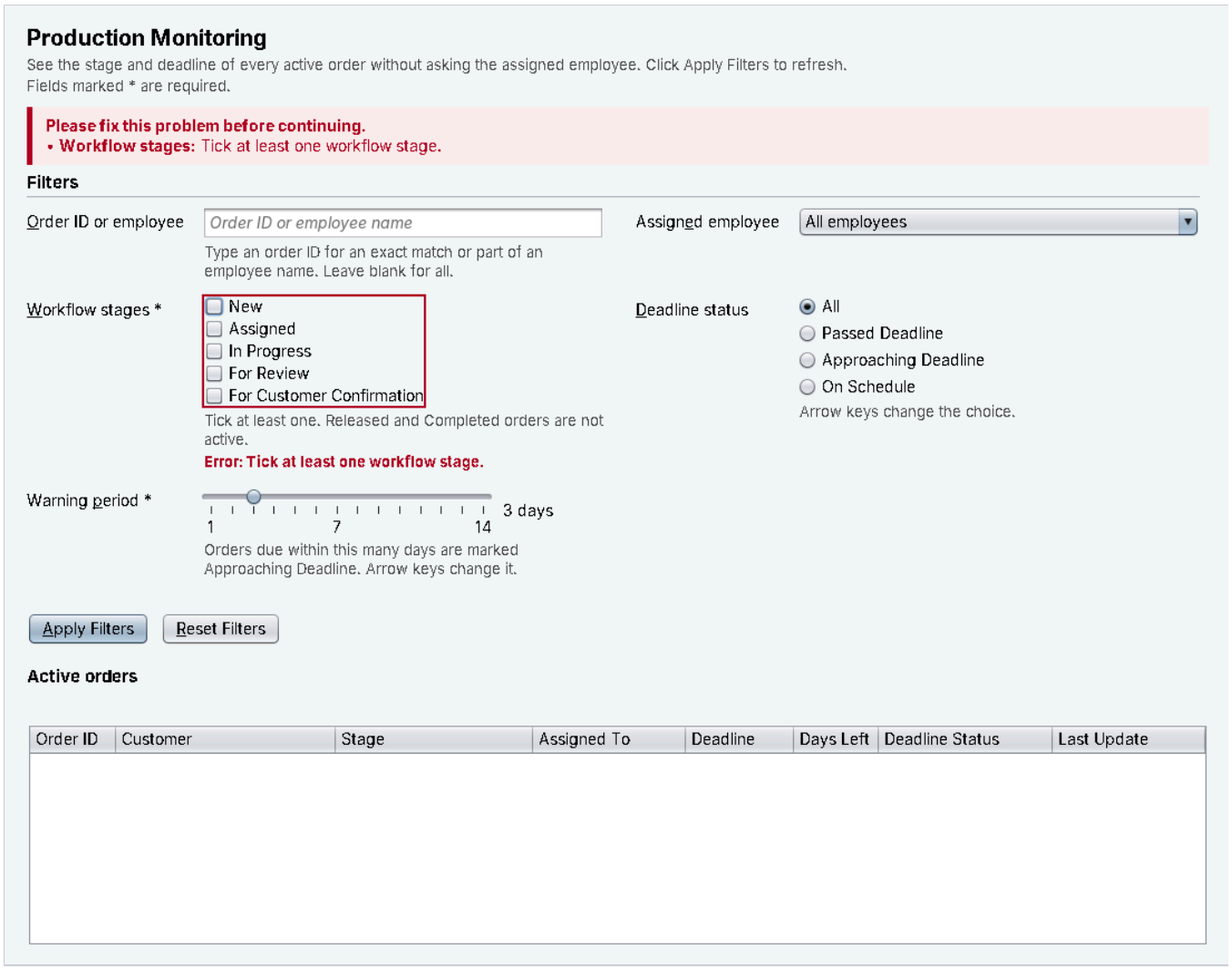
Orders per stage: New 2, Assigned 1, In Progress 2, For Review 1, For Customer Confirmation 1

Order ID	Customer	Stage	Assigned To	Deadline	Days Left	Deadline Status	Last Update
1002	Solid Rock Builders Inc.	Assigned	Jasmine Cruz	2026-10-26	-1	Passed Deadline	2026-10-20 15:20
1005	Kristine Ocampo	For Customer Confirma...	Paolo Reyes	2026-10-28	1	Approaching Deadline	2026-10-25 17:45
1003	Engr. Dennis Tan	New	Not assigned	2026-10-29	2	Approaching Deadline	2026-10-27 09:40
1009	Juan Dela Cruz	New	Not assigned	2026-10-31	4	On Schedule	2026-10-27 09:00
1001	Arch. Lea Navarro	In Progress	Paolo Reyes	2026-11-01	5	On Schedule	2026-10-24 11:05
1008	Engr. Dennis Tan	In Progress	Jasmine Cruz	2026-11-04	8	On Schedule	2026-10-22 16:00
1004	Bayview Construction Corp.	For Review	Mark Delos Santos	2026-11-08	12	On Schedule	2026-10-15 10:00

All active orders with deadline indicators



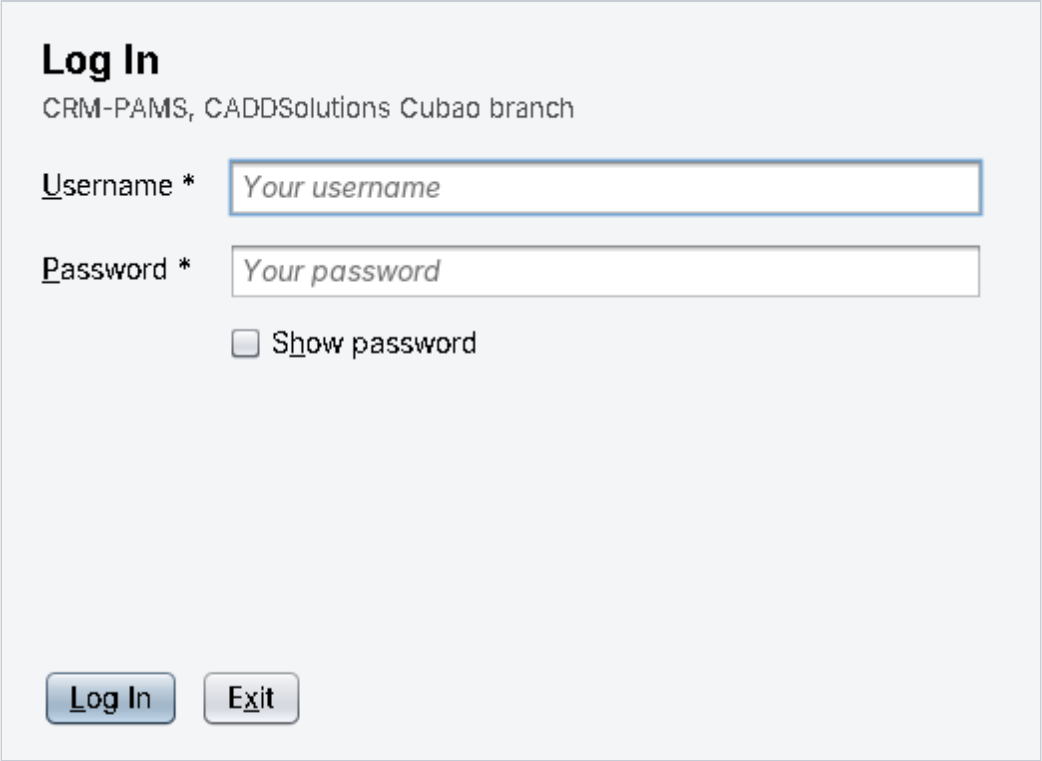
Approaching Deadline only, with a 7-day warning period



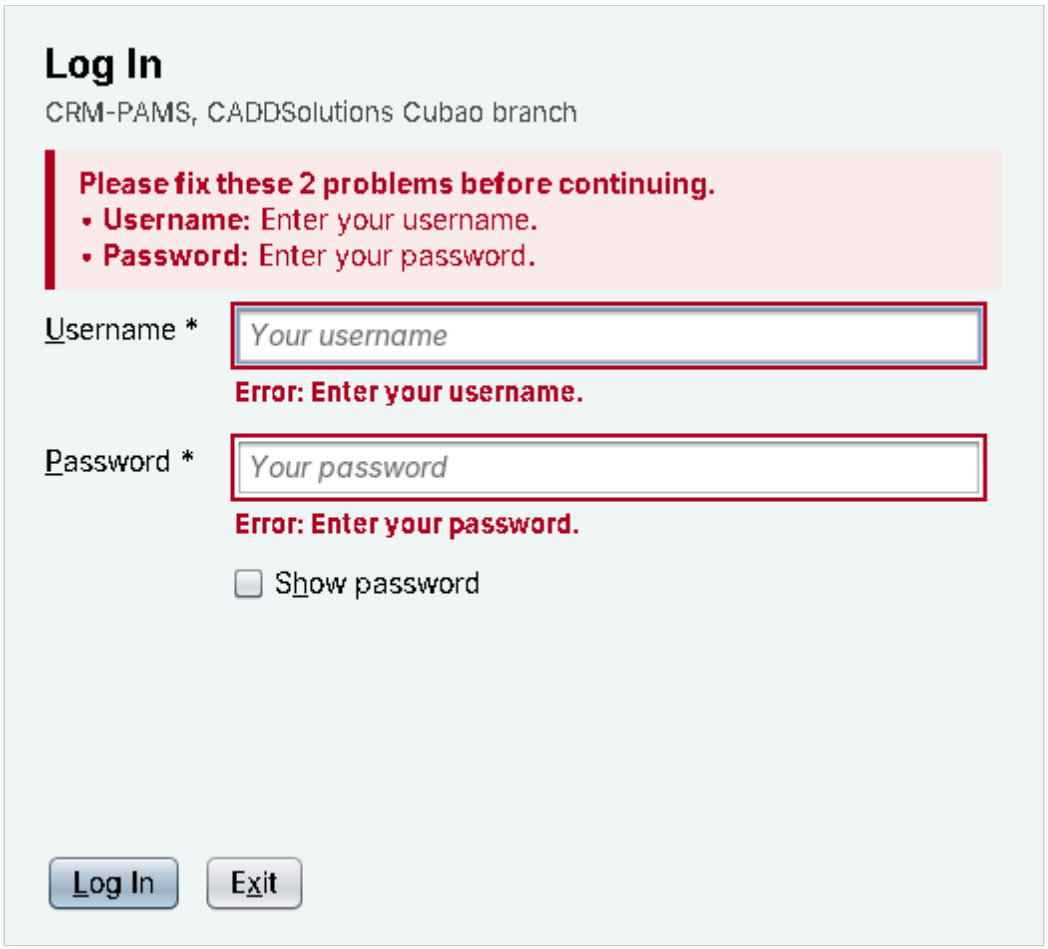
Production Monitoring, error state

6. Login Form and Main Window

The Login form (IPO 1.1) uses the same layout and validation. A wrong password and an unknown username show the same message, so the form does not reveal which usernames exist. Every login, failed login, and logout is written to the activity log. After login, the main window shows only the menu items of the user’s role (Chapter 2, Figures 2 to 4).



Login form



Login form, error state



Main window of the Operations Manager

7. Testing

The validation rules and services are covered by JUnit 6 tests, following the black box and white box approach in Chapter 3. The tests use an in-memory copy of the tables, so they run without a MySQL server.

Test class	What it checks
`FieldRulesTest`	Every rule with valid input, each invalid case, and boundary values (minimum and maximum lengths, deadline equal to today, end date equal to start date, warning period 1 and 14)
`PasswordHasherTest`	Correct and wrong passwords, random salt, 600,000 iterations by default
`DeadlineFlagTest`	Each branch of the deadline indicator rule
`LikeEscaperTest`	Percent signs and underscores are matched literally
`OrderIntakeServiceTest`	Saving a new customer and order together, reusing an existing customer, duplicate warning, closed orders not counted as duplicates, invalid input saving nothing, reference file name and location, role check
`RegistrationServiceTest`	Hashed password, position rule, unique username, matching passwords, role check
`SearchAndMonitoringServiceTest`	Text search, closed-status rule, date range, deadline flags, stage rule, role checks
`AuthServiceTest`	Each login outcome: missing input, wrong password, unknown user, deactivated account, success
`FormBehaviorTest`	Each form as a user would use it: invalid submit shows errors and saves nothing, valid submit saves, typing filters, at least 10 focusable controls per form, a label and accessible name for every input

Result of the last run: 63 tests, all passed.

8. Directory Structure

```
crm-pams/
|-- docs/
|   \-- form-controls/
|       |-- Form_Controls_Documentation.md
|       |-- TEAM_TASKS.md
|       \-- demo-video-script.md
|-- sql/
|   |-- schema.sql
|   \-- seed.sql
|-- src/
|   |-- main/
|   |   \-- java/
|   |       \-- crmpams/
|   |           |-- dao/
|   |           |   |-- jdbc/
|   |           |   |   |-- JdbcAccountDao.java
|   |           |   |   |-- JdbcAuditDao.java
|   |           |   |   |-- JdbcCustomerDao.java
|   |           |   |   |-- JdbcOrderDao.java
|   |           |   |   |-- JdbcReferenceDataDao.java
|   |           |   |   \-- JdbcSupport.java
|   |           |   |-- AccountDao.java
|   |           |   |-- AuditDao.java
|   |           |   |-- CustomerDao.java
|   |           |   |-- OrderDao.java
|   |           |   \-- ReferenceDataDao.java
|   |           |-- db/
|   |           |   |-- Database.java
|   |           |   |-- LikeEscaper.java
|   |           |   |-- SqlWork.java
|   |           |   \-- TransactionRunner.java
|   |           |-- model/
|   |           |   |-- CurrentUser.java
|   |           |   |-- Customer.java
|   |           |   |-- DeadlineFlag.java
|   |           |   |-- EmployeeRequest.java
|   |           |   |-- LookupItem.java
|   |           |   |-- MonitoringFilter.java
|   |           |   |-- MonitoringRow.java
|   |           |   |-- NewOrderRecord.java
|   |           |   |-- OrderRequest.java
|   |           |   |-- OrderSearchCriteria.java
|   |           |   |-- OrderSummary.java
|   |           |   |-- Role.java
|   |           |   |-- UrgencyLevel.java
|   |           |   |-- UserAccount.java
|   |           |   \-- WorkflowStages.java
|   |           |-- security/
|   |           |   |-- AccessControl.java
|   |           |   |-- AccessDeniedException.java
|   |           |   \-- PasswordHasher.java
|   |           |-- service/
|   |           |   |-- AuthService.java
|   |           |   |-- LoginResult.java
|   |           |   |-- MonitoringOptions.java
|   |           |   |-- MonitoringResult.java
|   |           |   |-- MonitoringService.java
|   |           |   |-- OrderFormData.java
|   |           |   |-- OrderIntakeService.java
|   |           |   |-- RegistrationService.java
|   |           |   |-- SaveOrderResult.java
|   |           |   |-- SaveResult.java
|   |           |   |-- SearchOptions.java
|   |           |   |-- SearchResult.java
|   |           |   \-- SearchService.java
|   |           |-- ui/
|   |           |   |-- components/
|   |           |   |   |-- DateSpinner.java
|   |           |   |   |-- FormPanel.java
|   |           |   |   |-- HintPainter.java
|   |           |   |   |-- HintPasswordField.java
|   |           |   |   |-- HintTextArea.java
|   |           |   |   |-- HintTextField.java
|   |           |   |   |-- InputFilters.java
|   |           |   |   |-- OrderedFocusPolicy.java
|   |           |   |   |-- RadioGroup.java
|   |           |   |   |-- RowTableModel.java
|   |           |   |   \-- ToggleSwitch.java
|   |           |   |-- forms/
|   |           |   |   |-- LoginForm.java
|   |           |   |   |-- MonitoringFilterForm.java
|   |           |   |   |-- NewOrderForm.java
|   |           |   |   |-- RegisterEmployeeForm.java
|   |           |   |   \-- SearchOrdersForm.java
|   |           |   |-- Dialogs.java
|   |           |   |-- MainWindow.java
|   |           |   |-- UiTasks.java
|   |           |   \-- UiTheme.java
|   |           |-- validation/
|   |           |   |-- FieldError.java
|   |           |   |-- FieldRules.java
|   |           |   |-- Fields.java
|   |           |   \-- ValidationResult.java
|   |           |-- App.java
|   |           \-- AppContext.java
|   \-- test/
|       \-- java/
|           \-- crmpams/
|               |-- db/
|               |   \-- LikeEscaperTest.java
|               |-- model/
|               |   \-- DeadlineFlagTest.java
|               |-- security/
|               |   \-- PasswordHasherTest.java
|               |-- service/
|               |   |-- AuthServiceTest.java
|               |   |-- OrderIntakeServiceTest.java
|               |   |-- RegistrationServiceTest.java
|               |   \-- SearchAndMonitoringServiceTest.java
|               |-- testsupport/
|               |   |-- InMemoryStore.java
|               |   \-- TestFixtures.java
|               |-- tools/
|               |   \-- FormScreenshotTool.java
|               |-- ui/
|               |   \-- forms/
|               |       |-- FormBehaviorTest.java
|               |       \-- FormDriver.java
|               \-- validation/
|                   \-- FieldRulesTest.java
|-- README.md
|-- db.properties.example
\-- pom.xml
```

Project directory structure

9. Items to Confirm Before Submission

These are design decisions made while building the forms. They do not change the six modules, but the group should confirm them.

1. **Table and column names.** The database schema in `sql/schema.sql` was rebuilt from the written descriptions in Chapters 1 to 3. Compare it with Figure 89 (Entity Relationship Diagram) and rename columns to match the ERD where they differ.
2. **Email columns.** Email is stored for customers and employees as an optional field, based on “customer name and contact details” in Chapter 1. Remove it if the ERD has no email column.
3. **Project description is required** (5 to 500 characters), because the duplicate check compares descriptions. Chapter 1 names only customer, project type, and deadline as required.
4. **Photoshop files (psd)** were added to the accepted reference file types because the interview names PDF, AutoCAD, and Photoshop as the files clients send.
5. **Urgency levels** in the sample data are Low (1), Normal (2), and High (3). Chapter 3 uses Normal 2 and High 3; Low was added as the lowest level.
6. **Contact number formats** accept Philippine mobile numbers and Metro Manila landlines only.